

Customer Shopping Behavior Analysis

An end-to-end analysis of 3,900 retail customers — quantifying what drives revenue, discounts, subscriptions, and loyalty across categories, seasons, and channels, with prioritized recommendations for marketing and merchandising.

PYTHON · POSTGRESQL · POWER BI

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Executive Summary

This report analyzes the purchasing behavior of **3,900 retail customers** across four product categories, all 50 US states, and four seasons, using a structured Python - PostgreSQL - Power BI pipeline. The analysis answers ten business questions covering revenue concentration, demographics, discount effectiveness, loyalty, and operations. Headline results:

\$233,081 TOTAL REVENUE	3,900 CUSTOMERS	\$59.76 AVG ORDER VALUE
3.75 / 5.0 AVG REVIEW RATING	27% SUBSCRIPTION RATE	43% PURCHASES DISCOUNTED

- **Clothing dominates the assortment:** \$104,264 revenue (44.7% of total) from 1,737 orders; the four categories are strongly concentrated at the top.
- **Revenue is customer-count driven, not spend driven:** male customers generate 67.7% of revenue because they are 68% of the base; per-customer spend is nearly identical (\$59.54 vs \$60.25).
- **Discounts do not lift basket size:** discounted orders average \$59.28 versus \$60.13 without a discount, yet 43% of purchases carry one.
- **Subscription value is frequency, not basket:** subscribers (27% of customers) spend \$59.49 per order versus \$59.87 for non-subscribers - the program is not yet a spend lever.
- **The base is loyal but thin at the top of the funnel:** 79.9% of customers qualify as Loyal (11+ prior purchases); only 83 customers (2.1%) are New.
- **Demand is season-proof:** all four seasons finish within a 7.6% revenue band, and no age cohort (18-70) is materially weaker.

1. Background and Objectives

A retail company wants a clearer picture of how its customers buy: which segments and products drive revenue, whether discounts and subscriptions change spending behavior, and where marketing and merchandising effort should concentrate. The overarching business question, as framed in the project brief, is: *"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"*

The analysis was scoped around ten business questions:

revenue by gender; discount users who still outspend the average; top-rated products; shipping-type spend comparison; subscriber economics; discount-dependent products; loyalty segmentation; best-selling items per category; repeat-buyer subscription behavior; and revenue by age group.

2. Data and Methodology

Source. Customer Shopping Trends Dataset (Kaggle, Sourav Banerjee) - a synthetic retail dataset of 3,900 rows and 18 columns covering demographics, purchase details, and shopping behavior. Each row represents one customer snapshot, not a timestamped transaction.

Pipeline. The raw CSV was cleaned and enriched in Python (pandas), loaded into PostgreSQL, queried with SQL for the ten business questions, and visualized in an interactive Power BI dashboard. Preparation steps:

- Imputed 37 missing Review Rating values (1.0% of rows) with the median rating of the customer's product category, preserving category-level rating structure.
- Standardized column names to snake_case for SQL compatibility.
- Engineered **age_group** (four equal-sized quartiles: Young Adult 18-31, Adult 32-44, Middle-aged 45-57, Senior 58-70) and **purchase_frequency_days** (stated purchase cadence converted to approximate days between purchases).
- Dropped **promo_code_used** after verifying it is identical to **discount_applied** in 100% of rows (redundant column).

Limitations. The dataset is synthetic and evenly distributed by design, so seasonal and geographic variation is muted by construction; there are no transaction dates, so time-series and trend analysis are out of scope. Findings should be read as a method demonstration and hypothesis set for real transactional data, not as market facts.

3. Findings

3.1 Revenue is concentrated in Clothing (F1)

Clothing generates nearly half of all revenue, followed by Accessories; Outerwear is a distant fourth. Order counts follow the same order, so the concentration reflects demand volume rather than price differences - average order value sits in a narrow \$57-\$62 band in every category.

Category	Orders	Revenue	Share	Avg order
Clothing	1737	\$104,264	44.7%	\$60.03
Accessories	1240	\$74,200	31.8%	\$59.84
Footwear	599	\$36,093	15.5%	\$60.26
Outerwear	324	\$18,524	7.9%	\$57.17

Table 1. Revenue and order mix by category.

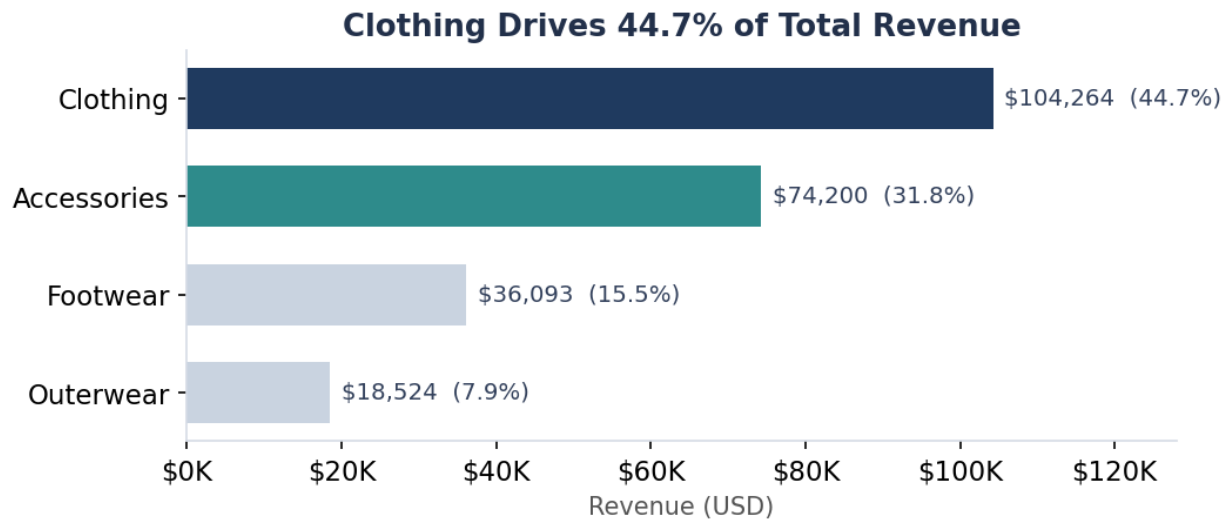


Figure 1. Clothing drives 44.7% of total revenue.

3.2 The gender revenue gap is a customer-count gap (F2)

Male customers account for \$157,890 (67.7% of revenue) against \$75,191 for female customers. However, average spend per customer is statistically indistinguishable (\$59.54 vs \$60.25 - female customers actually edge ahead). The revenue gap is therefore explained almost entirely by base size: 2,652 male versus 1,248 female customers. Growing the female base, not lifting anyone's basket, is the revenue opportunity.

Gender	Customers	Revenue	Avg per customer
Male	2652	\$157,890	\$59.54
Female	1248	\$75,191	\$60.25
Total	3900	\$233,081	\$59.76

Table 2. Revenue by gender (business question 1).

3.3 Young adults lead a flat age curve (F3)

Young Adults (18-31) are the top-revenue cohort at \$62,143, but the spread across all four cohorts is only 11%. Spending does not decay with age in this base - messaging can stay broad, with a light tilt toward younger shoppers.

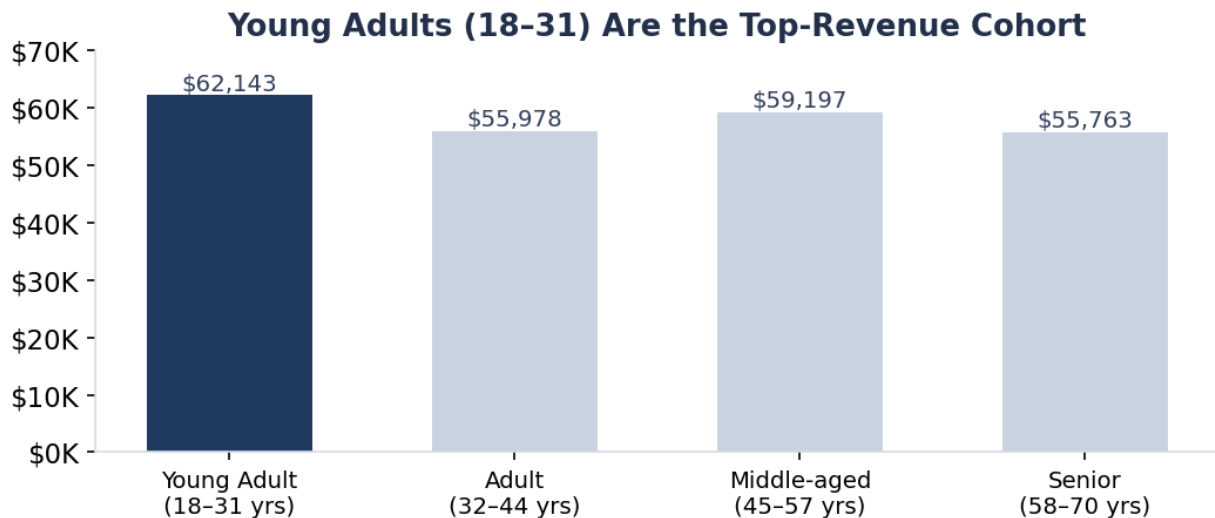


Figure 2. Revenue by age quartile (business question 10).

3.4 Subscribers do not spend more per order (F4)

Only 1,053 customers (27.0%) subscribe. Subscriber average order value is \$59.49 versus \$59.87 for non-subscribers - a 0.6% gap with no practical significance. Repeat buyers (6+ prior purchases) subscribe at 27.6% versus the 27.0% baseline, so purchase history alone is also not converting customers into subscribers. The subscription program currently monetizes frequency and retention, not basket size; its benefits should be framed (and measured) accordingly.

Segment	Customers	Avg spend	Total revenue
Subscribed	1053	\$59.49	\$62,645
Not subscribed	2847	\$59.87	\$170,436

Table 3. Subscriber vs non-subscriber economics (business questions 5 and 9).

Only 1 in 4 Customers Subscribes

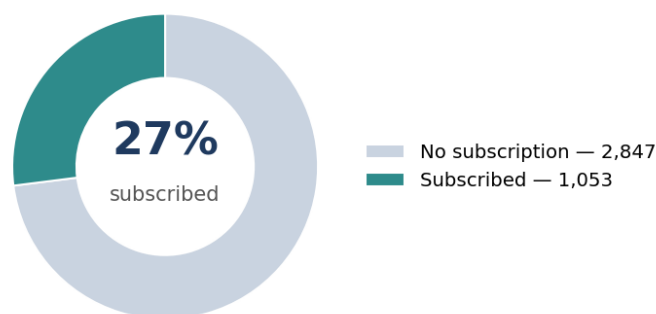


Figure 3. Subscription penetration (27% of 3,900 customers).

3.5 Discounts are widespread but do not grow baskets (F5)

Discounts were applied to 43% of purchases (1,677), yet discounted orders average \$59.28 against \$60.13 for undiscounted orders - customers do not add more to the basket when a discount applies. Two nuances matter for policy. First, 839 customers used a discount and still spent at or above the \$59.76 average, together worth \$66,942 - a "smart shopper" segment that combines price sensitivity with willingness to spend. Second, discount attachment is concentrated on specific items: Hat (50.0%), Sneakers (49.7%), Coat (49.1%), Sweater (48.2%) and Pants (47.4%) top the list, effectively halving margin on core products without a basket-size payoff.

Measure	With discount	Without discount
Share of purchases	43.0% (1,677)	57.0% (2,223)
Average order value	\$59.28	\$60.13

Table 4. Discount impact on order value (business questions 2 and 6).

3.6 The customer base is loyal but rarely refreshes (F6)

Segmenting by purchase history produces 3,116 Loyal customers (11+ prior purchases, 79.9%), 701 Returning (2-10) and just 83 New (first purchase, 2.1%). The base is deep but the acquisition funnel is thin: revenue depends on retaining veterans, while almost no first-time buyers are entering to replace eventual churn. Retention and acquisition deserve separate, explicit targets.

Segment	Definition (prior purchases)	Customers	Share
Loyal	11 or more	3116	79.9%
Returning	2 - 10	701	18.0%
New	1	83	2.1%

Table 5. Loyalty segmentation (business question 7).

3.7 Product performance: ratings and best sellers point the same way (F7)

The best-rated products are Gloves (3.86), Sandals (3.84) and Boots (3.82) - practical accessories and footwear, while the revenue leaders are core clothing staples: Blouse (\$10,410), Shirt (\$10,332) and Dress (\$10,320). Within each category the top items are clear (Table 6). Campaigns that pair high satisfaction items with high-revenue staples (for example, gloves with winter clothing) can lift both.

Category	Top items (rank)	Orders	Rank
Accessories	Jewelry	171	#1
Accessories	Belt	161	#2
Accessories	Sunglasses	161	#2
Accessories	Scarf	157	#3
Clothing	Blouse	171	#1
Clothing	Pants	171	#1
Clothing	Shirt	169	#2

Category	Top items (rank)	Orders	Rank
Clothing	Dress	166	#3
Footwear	Sandals	160	#1
Footwear	Shoes	150	#2
Footwear	Sneakers	145	#3
Outerwear	Jacket	163	#1
Outerwear	Coat	161	#2

Table 6. Best-selling items within each category (business question 8).

3.8 Operations: flat seasons, express correlates with higher spend (F8)

Seasonal revenue sits in a 7.6% band (Fall \$60,018 to Summer \$55,777), so inventory and staffing can be planned level across the year with modest fall emphasis. Customers choosing Express shipping average \$60.48 per order versus \$58.46 for Standard - express buyers correlate with a 3.5% higher basket. Payment preferences are balanced (PayPal leads at 677 of 3,900), requiring no single-rail strategy.

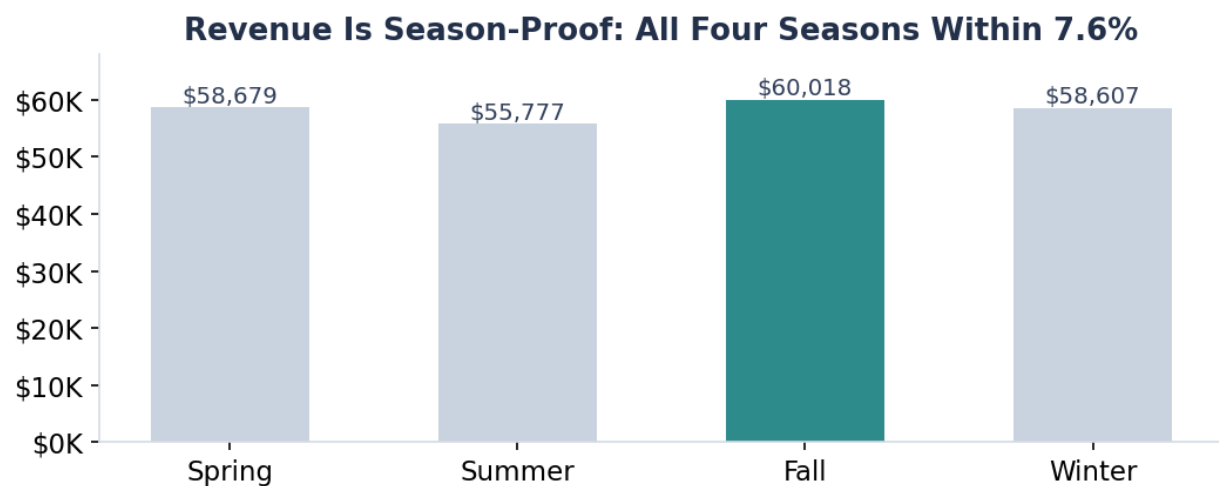


Figure 4. Revenue by season (business questions 4 and seasonal scope).

4. Recommendations

R1 - Reframe subscriptions around frequency benefits (*High priority, Marketing*)

Subscribers do not spend more per order (F4), and repeat buyers subscribe at nearly the base rate. Target the 2,518 repeat buyers who are not subscribed with frequency-based perks (free shipping, early access) rather than discount promises; measure success in purchase cadence and retention, not AOV.

R2 - Cap blanket discounts on the five most discount-dependent items (*High priority, Category management*)

Hat, Sneakers, Coat, Sweater and Pants carry discounts on 47-50% of purchases with no basket-size uplift (F5). Shift spend from item-wide discounts to threshold offers (for example, spend \$70, save 10%) designed to grow the basket, and A/B test margin recovery on these five items.

R3 - Launch a female-customer acquisition campaign (*Medium priority, Marketing*)

The revenue gap is a base-size gap, not a spend gap (F2); female customers already spend marginally more per head. Acquisition - not basket persuasion - closes the \$83K headroom implied by base parity.

R4 - Pair top-rated items with revenue staples in campaigns (*Medium priority, Merchandising*)

Gloves, Sandals and Boots lead satisfaction while Blouse, Shirt and Dress lead revenue (F7). Bundle or cross-merchandise satisfaction leaders with staples to convert goodwill into order value.

R5 - Offer express-shipping upgrades at checkout (*Lower priority, Operations*)

Express orders run 3.5% larger (F8). A low-cost upgrade prompt nudges Standard buyers toward the behavior already associated with higher baskets, and shipping revenue partially offsets the perk.

5. Deliverables and Dashboard

The analysis ships with an interactive Power BI dashboard (customer_behavior_dashboard.pbix) connected to the PostgreSQL database. It presents KPI cards, subscription share, and revenue and customer counts by category and age group, with slicers for gender, category, subscription status, and shipping type - allowing every finding in Section 3 to be reproduced interactively.

Measure design. Every count visual is driven by explicit DAX measures rather than implicit column aggregation: Number of Customers = COUNT(customer_id), Average Purchase Amount = AVERAGE(purchase_amount), and Average Review Rating = AVERAGE(review_rating). Customer-count charts therefore plot true counts (for example, the subscription split of 2,847 / 1,053) instead of summed identifier values - a defect identified and corrected during a field-binding audit of every visual.

Repository artifacts. Cleaning notebook (Python), SQL business-question file, Power BI dashboard, executive presentation (12 slides), and this report - with a README documenting reproduction steps.

Appendix A. Result Tables by Business Question

#	Business question	Result
Q1	Revenue by gender	Male \$157,890; Female \$75,191
Q2	Discount users at/above avg spend	839 customers, \$66,942 combined

#	Business question	Result
Q3	Top 5 products by avg rating	Gloves 3.86; Sandals 3.84; Boots 3.82; Hat 3.80; Handbag 3.78
Q4	Express vs Standard avg spend	Express \$60.48; Standard \$58.46
Q5	Subscriber vs non-subscriber	1,053 subs, \$59.49 avg; 2,847 non-subs, \$59.87 avg
Q6	Most discount-dependent products	Hat 50.0%; Sneakers 49.7%; Coat 49.1%; Sweater 48.2%; Pants 47.4%
Q7	Loyalty segments	Loyal 3,116; Returning 701; New 83
Q8	Top 3 items per category	See Table 6
Q9	Repeat buyers and subscription	2,518 non-subscribed vs 958 subscribed (27.6% vs 27.0% base)
Q10	Revenue by age group	Young Adult \$62,143; Middle-aged \$59,197; Adult \$55,978; Senior \$55,763

Table A1. Consolidated SQL results (all figures computed from the cleaned dataset).

Appendix B. Data Dictionary (cleaned table)

Column	Type	Description
customer_id	Integer	Unique customer identifier
age / age_group	Integer / Text	Customer age; quartile group (Young Adult, Adult, Middle-aged, Senior)
gender	Text	Male or Female
item_purchased / category	Text	Product (25 items) and category (4)
purchase_amount	Integer	Purchase value in USD (20-100)
location	Text	US state (50)
size / color / season	Text	Size (S-XL), color (25), season (4)
review_rating	Decimal	Product rating, 2.5-5.0 (37 gaps imputed by category median)
subscription_status	Text	Subscription Yes/No
shipping_type	Text	6 shipping methods
discount_applied	Text	Yes/No (promo_code_used dropped as identical)
previous_purchases	Integer	Count of prior purchases (1-50)
payment_method	Text	6 payment methods
frequency_of_purchases / purchase_frequency_days	Text / Integer	Stated cadence and its approx. day equivalent

Table B1. Cleaned table schema loaded to PostgreSQL (database: customer_behavior, table: customer).